Diode models

Cheat sheet · Day 03 · Mon 26 Oct · pack E2

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
DIODE MODELS (T2, Wk 1-2)
V_T = kT_K/q  (k=1.38e-23, q=1.6e-19; T_K=273+C)  27C->25.875mV, 25C->25.72, 20C->25
I_D = I_S(e^{V_D/nV_T} - 1) ~ I_S e^{V_D/nV_T} (forward);  reverse: I_D ~ -I_S
V_D = nV_T ln(I_D/I_S)       dV_D = nV_T ln(I2/I1)   [I_S cancels]
1 decade = 2.303 nV_T  (n=1: ~60 mV; n=1.5: ~89 mV)
Two points: n = (V2-V1)/(V_T ln(I2/I1)); I_S = I1 e^{-V1/nV_T}; check with point 2
Parallel identical diodes, same I: V falls by nV_T ln2 (~18 mV)
TEMP: V_D falls 2.5 mV/C (fixed I); I_S doubles per 10C: x 2^{dT/10}; V_T ~ T
R_D = V_D/I_D (dc, static)   r_d = dV/dI = nV_T/I_D (ac/small-signal, v_d < ~5 mV)
r_av = dV/dI between two points -> 3rd model (V_K + r_av). Knee V_K: Ge 0.3, Si 0.7,
MODELS  ON:  1 ideal = short | 2 = V_D source | 3 = V_D + r_D   (+ on anode side)
        OFF: open circuit in ALL models. 3rd model: v_d = V_D + i_d r_D
ASSUME-VERIFY: 1 model  2 guess each diode (write it)  3 redraw+solve: i_d for ON,
  v_d = V_anode - V_cathode for OFF (OFF branch: no I, resistors drop 0, batteries co
  4 check ON: i_d > 0 ; OFF: v_d < V_D (ideal: < 0). Fail -> flip that diode, resolve
Tip: Thevenin everything except the diode -> i_d = (V_Th - V_D)/(R_Th + r_D)
PT Q2: 0.179 V.  PT Q3: I = 15.3 uA, Vo = 3.15 V (0.7 V model would make them OFF!)
```